

Boutique Shop Management System

System Architecture, Entity Relationships & Database Design Document

Introduction:

This document details the database architecture, design choices, business validation rules, and schemas for the Boutique Shop Management System backend. It is designed to act as a complete technical specification for database developers and system integrators. The database is built on a highly normalized schema with robust constraints to ensure absolute data consistency.

Core Database Design Decisions:

- **UUID Primary Keys:** All tables use universally unique identifiers (UUIDs) stored as 36-character strings (mapped to VARCHAR(36) in MySQL). This facilitates horizontal scalability, prevents ID guessing attacks, and allows offline clients (like mobile POS apps) to generate unique record keys without a database roundtrip.
- **Soft-Deletion Policy:** Rather than hard deleting entries (which damages financial histories and audit trails), the system utilizes soft deletion. Deleting a record sets `is_deleted = True` and records the timestamp in `deleted_at`. SQLAlchemy queries automatically filter out soft-deleted records by default.
- **State Delta Auditing:** Database mutations are logged to the `audit_logs` table. This stores a JSON-encoded snapshot of the row state before (`old_value`) and after (`new_value`) the update, along with the actor's ID, request route, IP address, and client details.

Business Rules & Constraints Enforced:

- **Non-Negative Stock:** Stock values are validated at both the API and database levels. Inventory levels cannot drop below zero. Stock additions and subtractions generate immutable history logs.
- **Valuation Math:** Stock purchase costs adjust a weighted `average_cost` inside the `inventory` table, calculating total financial inventory valuation (`average_cost * quantity`).
- **Sales-Profit Coupling:** Creating a sale automatically reduces inventory levels, copies the item's purchase unit cost into `sale_items.unit_cost` (fixing the cost snapshot in case prices change later), calculates the net profit margin, and registers the cash flow in the active `capital` ledger.
- **Price Integrity:** Buying wholesale price cannot exceed POS selling price without explicitly overridden Administrator permissions.

Entity-Relationship Overview:

- **One-to-Many Relationships:** Categories to Products, Brands to Products, Users to Products (created by), Users to Sales (cashier), Customers to Sales, Sales to SaleItems, Suppliers to Products/Purchases, Purchases to PurchaseItems, ExpenseCategories to Expenses, and Users to AuditLogs.
- **Many-to-Many Relationships:** Users to Roles (via `user_roles`), and Roles to Permissions (via `role_permissions`).
- **One-to-One Relationships:** Products to Inventory.

1. Authentication & RBAC Modules

users Table

Holds credentials and profiles for cashiers, managers, and administrators.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique User identifier (UUID)
username	VARCHAR(50)	UNIQUE, INDEX, NOT NULL	Unique username for system login
email	VARCHAR(100)	UNIQUE, INDEX, NOT NULL	User email address
password_hash	VARCHAR(255)	NOT NULL	Bcrypt hashed password
first_name	VARCHAR(50)	NOT NULL	First name
last_name	VARCHAR(50)	NOT NULL	Last name
phone	VARCHAR(20)	Nullable	Contact number
address	VARCHAR(255)	Nullable	Home/office address
profile_image	VARCHAR(255)	Nullable	Path to profile picture
is_active	BOOLEAN	DEFAULT True, NOT NULL	Indicates if user account is enabled
last_login	DATETIME	Nullable	Timestamp of last successful login

roles Table

Predefined user authorization tiers.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique Role identifier (UUID)
name	VARCHAR(50)	UNIQUE, NOT NULL	Role name (Administrator, Manager, Cashier)
description	VARCHAR(255)	Nullable	Brief role capability description

permissions Table

Granular application capabilities checking.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique Permission identifier (UUID)
name	VARCHAR(100)	UNIQUE, NOT NULL	Unique permission string (e.g. manage_sales)
description	VARCHAR(255)	Nullable	Description of what it permits
module	VARCHAR(50)	NOT NULL	Associated module (sales, products, etc.)

user_roles Table (Association Table)

Establishes many-to-many joins between users and roles.

Column	Data Type	Constraints	Description
user_id	VARCHAR(36)	FOREIGN KEY(users.id), NOT NULL	Associated User UUID
role_id	VARCHAR(36)	FOREIGN KEY(roles.id), NOT NULL	Associated Role UUID

role_permissions Table (Association Table)

Establishes many-to-many joins between roles and permissions.

Column	Data Type	Constraints	Description
role_id	VARCHAR(36)	FOREIGN KEY(roles.id), NOT NULL	Associated Role UUID
permission_id	VARCHAR(36)	FOREIGN KEY(permissions.id), NOT NULL	Associated Permission UUID

2. Catalog & Inventory Modules

categories Table

Product categorization hierarchy.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique Category identifier (UUID)
name	VARCHAR(50)	UNIQUE, INDEX, NOT NULL	Category name (e.g. Dresses, Accessories)
description	VARCHAR(255)	Nullable	Category description

brands Table

Product designer/label categorization.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique Brand identifier (UUID)
name	VARCHAR(50)	UNIQUE, INDEX, NOT NULL	Brand label name (e.g. Nike, Zara)
description	VARCHAR(255)	Nullable	Brand description

products Table

Primary inventory item registry storing core item details and pricing information.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique Product identifier (UUID)
product_code	VARCHAR(20)	UNIQUE, INDEX, NOT NULL	Auto-generated PRD-XXXXXXX code
barcode	VARCHAR(50)	UNIQUE, INDEX, Nullable	Literal barcode text scanned/stored
qr_code	VARCHAR(255)	UNIQUE, Nullable	Optional path/reference to QR code
name	VARCHAR(100)	INDEX, NOT NULL	Product name label
description	TEXT	Nullable	Long product description details
category_id	VARCHAR(36)	FOREIGN KEY(categories.id), NOT NULL	Classification category link
brand_id	VARCHAR(36)	FOREIGN KEY(brands.id), NOT NULL	Classification brand link
gender	VARCHAR(10)	DEFAULT 'unisex', NOT NULL	Target segment (men, women, kids, unisex)
color	VARCHAR(50)	Nullable	Product color attribute
size	VARCHAR(20)	Nullable	Product size attribute (S, M, L, XL, etc.)
buying_price	DECIMAL(10,2)	NOT NULL	Item wholesale buying cost
selling_price	DECIMAL(10,2)	NOT NULL	Item POS retail selling price
quantity	INTEGER	DEFAULT 0, NOT NULL	Current inventory level
minimum_stock	INTEGER	DEFAULT 10, NOT NULL	Low stock indicator threshold level
supplier_id	VARCHAR(36)	FOREIGN KEY(suppliers.id), Nullable	Associated default supplier ID
image	VARCHAR(255)	Nullable	Link/path to product image
status	VARCHAR(20)	DEFAULT 'active', NOT NULL	Product state (active, inactive, discontinued)
created_by	VARCHAR(36)	FOREIGN KEY(users.id), NOT NULL	Admin/Staff user UUID who added it

inventory Table

Monitors available, reserved, and average costing valuation for products.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique Inventory identifier (UUID)
product_id	VARCHAR(36)	FOREIGN KEY(products.id), UNIQUE, NOT NULL	Associated Product UUID
quantity	INTEGER	DEFAULT 0, NOT NULL	Gross stock level count
reserved_quantity	INTEGER	DEFAULT 0, NOT NULL	Reserved stock for pending checkouts
available_quantity	INTEGER	DEFAULT 0, NOT NULL	Net sellable count (quantity - reserved)
average_cost	DECIMAL(10,2)	DEFAULT 0.00	Weighted average price of purchased items
total_value	DECIMAL(10,2)	DEFAULT 0.00	Total inventory financial asset valuation

stock_movements Table

Comprehensive inventory transaction logging table.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique StockMovement identifier (UUID)
inventory_id	VARCHAR(36)	FOREIGN KEY(inventory.id), NOT NULL	Associated Inventory UUID
product_id	VARCHAR(36)	FOREIGN KEY(products.id), NOT NULL	Associated Product UUID
movement_type	VARCHAR(20)	NOT NULL	Log classification (stock_in, stock_out, adjustment)
quantity	INTEGER	NOT NULL	Quantity size of stock shift
previous_quantity	INTEGER	NOT NULL	Stock level prior to transaction
new_quantity	INTEGER	NOT NULL	Updated stock level post transaction
reference_type	VARCHAR(50)	Nullable	Audit source module (sale, purchase, adjustment)
reference_id	VARCHAR(36)	Nullable	Audit source module identifier link
reason	VARCHAR(255)	Nullable	Short descriptive text of movement
notes	TEXT	Nullable	Extended notes
performed_by	VARCHAR(36)	FOREIGN KEY(users.id), NOT NULL	Staff user UUID who made adjustment

3. Sales & POS Modules

customers Table

Customer database recording profiles and loyalty reward program balances.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique Customer identifier (UUID)
name	VARCHAR(100)	INDEX, NOT NULL	Customer full name
phone	VARCHAR(20)	NOT NULL	Contact phone number
email	VARCHAR(100)	Nullable	Email address
address	VARCHAR(255)	Nullable	Delivery/home address
gender	VARCHAR(10)	Nullable	Gender options (male, female, other)
birthday	DATE	Nullable	Birthday for tracking age & birthday promotions
image	VARCHAR(255)	Nullable	Path to profile picture
loyalty_points	INTEGER	DEFAULT 0, NOT NULL	Loyalty system accumulated points
is_active	BOOLEAN	DEFAULT True, NOT NULL	Account status flag

sales Table

Financial headers for POS retail transactions.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique Sale identifier (UUID)
invoice_number	VARCHAR(20)	UNIQUE, INDEX, NOT NULL	Unique invoice reference number
receipt_number	VARCHAR(20)	UNIQUE, NOT NULL	Unique receipt voucher string
customer_id	VARCHAR(36)	FOREIGN KEY(customers.id), Nullable	Customer UUID associated with sale
subtotal	DECIMAL(10,2)	DEFAULT 0.00	Gross purchase amount sum
discount	DECIMAL(10,2)	DEFAULT 0.00	Invoice deduction discount amount
tax	DECIMAL(10,2)	DEFAULT 0.00	Calculated tax size (18% by default)
total	DECIMAL(10,2)	DEFAULT 0.00	Net payable invoice sum
payment_method	VARCHAR(50)	NOT NULL	Payment type (cash, card, mobile_money, bank_transfer)
cash_received	DECIMAL(10,2)	DEFAULT 0.00	Cash amount handed over by customer
change	DECIMAL(10,2)	DEFAULT 0.00	Calculated return change amount
total_cost	DECIMAL(10,2)	DEFAULT 0.00	Aggregate item buying cost total for reporting
total_profit	DECIMAL(10,2)	DEFAULT 0.00	Net profit margin total
status	VARCHAR(20)	DEFAULT 'completed', NOT NULL	Voucher state (pending, completed, cancelled, refunded)
sale_date	DATETIME	DEFAULT utcnow, NOT NULL	Timestamp of point of sale completion
notes	TEXT	Nullable	Sale checkout notes
cashier_id	VARCHAR(36)	FOREIGN KEY(users.id), NOT NULL	Cashier/Staff user UUID who made checkout

sale_items Table

Itemized product checkout lines linked to sales.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique SaleItem identifier (UUID)
sale_id	VARCHAR(36)	FOREIGN KEY(sales.id), NOT NULL	Parent Sale UUID link
product_id	VARCHAR(36)	FOREIGN KEY(products.id), NOT NULL	Purchased Product UUID link
quantity	INTEGER	NOT NULL	Volume bought
unit_price	DECIMAL(10,2)	NOT NULL	Unit price at POS time
discount	DECIMAL(10,2)	DEFAULT 0.00	Unit discount deduction applied
subtotal	DECIMAL(10,2)	DEFAULT 0.00	Subtotal amount (price * qty - discount)
unit_cost	DECIMAL(10,2)	NOT NULL	Unit buying price at POS time (fixed for profit reports)

4. Purchasing & Expense Modules

suppliers Table

Registry of wholesale manufacturers/suppliers.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique Supplier identifier (UUID)
name	VARCHAR(100)	INDEX, NOT NULL	Supplier business name
contact_person	VARCHAR(100)	Nullable	Supplier contact person
phone	VARCHAR(20)	NOT NULL	Business contact phone number
email	VARCHAR(100)	Nullable	Business contact email
address	VARCHAR(255)	Nullable	Factory/head office physical address
tin_number	VARCHAR(50)	Nullable	Tax Identification Number
outstanding_balance	DECIMAL(10,2)	DEFAULT 0.00	Debt balance owed to the supplier
is_active	BOOLEAN	DEFAULT True, NOT NULL	Supplier active status flag

purchases Table

Financial headers for wholesale purchase orders made to restock catalog items.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique Purchase identifier (UUID)
purchase_number	VARCHAR(20)	UNIQUE, INDEX, NOT NULL	Unique purchase order document identifier
supplier_id	VARCHAR(36)	FOREIGN KEY(suppliers.id), NOT NULL	Associated Supplier UUID
subtotal	DECIMAL(10,2)	DEFAULT 0.00	Total items net cost sum
discount	DECIMAL(10,2)	DEFAULT 0.00	Total purchase order discount applied
tax	DECIMAL(10,2)	DEFAULT 0.00	Tax size calculated (18% by default)
total	DECIMAL(10,2)	DEFAULT 0.00	Total net cost payable
payment_method	VARCHAR(50)	NOT NULL	Payment type (cash, bank_transfer, credit)
paid_amount	DECIMAL(10,2)	DEFAULT 0.00	Amount paid immediately on order placement
balance	DECIMAL(10,2)	DEFAULT 0.00	Remaining unpaid balance (total - paid_amount)
purchase_date	DATETIME	DEFAULT utcnw, NOT NULL	Purchase order timestamp
status	VARCHAR(20)	DEFAULT 'pending', NOT NULL	Purchase state (pending, completed, cancelled)
notes	TEXT	Nullable	PO checkout notes
receipt_image	VARCHAR(255)	Nullable	Link/path to invoice PDF or image scan
created_by	VARCHAR(36)	FOREIGN KEY(users.id), NOT NULL	Staff user UUID who placed the PO

purchase_items Table

Itemized product checkout lines linked to purchases.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique PurchaseItem identifier (UUID)
purchase_id	VARCHAR(36)	FOREIGN KEY(purchases.id), NOT NULL	Parent Purchase UUID link
product_id	VARCHAR(36)	FOREIGN KEY(products.id), NOT NULL	Purchased Product UUID link
quantity	INTEGER	NOT NULL	Volume bought
buying_price	DECIMAL(10,2)	NOT NULL	Agreed unit buying cost
discount	DECIMAL(10,2)	DEFAULT 0.00	Unit discount applied
subtotal	DECIMAL(10,2)	DEFAULT 0.00	Subtotal cost (price * qty - discount)

expense_categories Table

Predefined operational expense classifications.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique Category identifier (UUID)
name	VARCHAR(50)	UNIQUE, INDEX, NOT NULL	Expense category name (Rent, Salary, etc.)
description	VARCHAR(255)	Nullable	Brief category explanation
is_recurring	BOOLEAN	DEFAULT False, NOT NULL	Indicates if categories are recurring monthly

expenses Table

Individual business operational expenditure records.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique Expense identifier (UUID)
name	VARCHAR(100)	NOT NULL	Short title description of the expense
description	TEXT	Nullable	Long text expense details
category_id	VARCHAR(36)	FOREIGN KEY(expense_categories.id), NOT NULL	Associated Category UUID link
amount	DECIMAL(10,2)	NOT NULL	Amount cost
expense_date	DATETIME	DEFAULT utctnow, NOT NULL	Transaction billing date
is_recurring	BOOLEAN	DEFAULT False, NOT NULL	Indicates if expense is recurring
recurring_month	INTEGER	Nullable	Target month value for recurring items (1-12)
receipt_image	VARCHAR(255)	Nullable	Link to image scan of payment receipt
notes	TEXT	Nullable	Expense notes
created_by	VARCHAR(36)	FOREIGN KEY(users.id), NOT NULL	Staff user UUID who created the expense entry

5. Capital, Settings, Auditing & Notifications

capital Table

Monitors current and cumulative capital ledger positions for financial modeling.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique Capital ledger ID (UUID)
beginning_capital	DECIMAL(12,2)	DEFAULT 0.00	Starting account balance at period start
current_capital	DECIMAL(12,2)	DEFAULT 0.00	Active liquidity balance
total_invested	DECIMAL(12,2)	DEFAULT 0.00	Aggregated capital investment injects
total_withdrawn	DECIMAL(12,2)	DEFAULT 0.00	Aggregated owner capital drawings
period_start	DATETIME	DEFAULT utcnow, NOT NULL	Financial calendar period start
period_end	DATETIME	Nullable	Financial calendar period end
is_active	BOOLEAN	DEFAULT True, NOT NULL	Indicates active financial period

settings Table

System configuration metadata parameters (typically single-row tables).

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique Settings row ID (UUID)
shop_name	VARCHAR(100)	NOT NULL	Retail shop commercial name
logo	VARCHAR(255)	Nullable	File path or URL to shop logo
address	VARCHAR(255)	Nullable	Retail shop commercial physical address
phone	VARCHAR(20)	Nullable	Shop contact telephone number
email	VARCHAR(100)	Nullable	Shop support email address
currency	VARCHAR(3)	DEFAULT 'USD', NOT NULL	Global base currency choice (e.g. USD, EUR)
currency_symbol	VARCHAR(5)	DEFAULT '\$', NOT NULL	Global base currency printable symbol
tax_percentage	INTEGER	DEFAULT 18, NOT NULL	Global default tax percentage rate (integer value)
receipt_header	TEXT	Nullable	Custom print header text on physical receipts
receipt_footer	TEXT	Nullable	Custom print footer text on physical receipts
low_stock_limit	INTEGER	DEFAULT 10, NOT NULL	Default global inventory low alert threshold
dark_mode	BOOLEAN	DEFAULT False, NOT NULL	Default interface display theme setting
tin_number	VARCHAR(50)	Nullable	Tax Identification Number for receipt billing

notifications Table

Stores alerts, notices, and integration logs for FCM Push messaging.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique Notification identifier (UUID)
user_id	VARCHAR(36)	FOREIGN KEY(users.id), NOT NULL	Recipient User UUID link
title	VARCHAR(100)	NOT NULL	Notification title header text
message	TEXT	NOT NULL	Long message details
notification_type	VARCHAR(50)	NOT NULL	Category (low_stock, out_of_stock, expense_reminder, etc.)
reference_type	VARCHAR(50)	Nullable	Optional linked system module category
reference_id	VARCHAR(36)	Nullable	Optional linked system record ID
is_read	BOOLEAN	DEFAULT False, NOT NULL	Boolean indicating read status
read_at	DATETIME	Nullable	Timestamp when recipient marked as read
push_sent	BOOLEAN	DEFAULT False, NOT NULL	Boolean indicating Firebase push execution
push_sent_at	DATETIME	Nullable	Timestamp when push notification was dispatched

audit_logs Table

System action trace logs capturing row-level mutation changes.

Column	Data Type	Constraints	Description
id	VARCHAR(36)	PRIMARY KEY, NOT NULL	Unique AuditLog identifier (UUID)
user_id	VARCHAR(36)	FOREIGN KEY(users.id), NOT NULL	UUID of acting User
username	VARCHAR(50)	NOT NULL	Snapshot username of actor
action	VARCHAR(50)	NOT NULL	Classification of action (create, update, delete, login)
entity_type	VARCHAR(50)	NOT NULL	Target resource name (product, sale, user, etc.)
entity_id	VARCHAR(36)	Nullable	Target resource primary key value
old_value	TEXT	Nullable	JSON text representation of prior record state
new_value	TEXT	Nullable	JSON text representation of updated record state
ip_address	VARCHAR(45)	Nullable	IP Address of request client
user_agent	VARCHAR(255)	Nullable	Browser user agent metadata header
request_method	VARCHAR(10)	Nullable	HTTP Verb of request (POST, PUT, DELETE)
request_path	VARCHAR(255)	Nullable	HTTP Request routing path
status	VARCHAR(20)	DEFAULT 'success', NOT NULL	Status of log operation (success, failure)
error_message	TEXT	Nullable	Standard stack trace or error response message if failed